Literature Review

EV Battery Health Prediction Using Limited and Noisy Sensor Data

1. Introduction

Accurate estimation of lithium-ion battery condition is central to electric vehicle (EV) safety, maintenance planning, and battery management. Two quantities are of primary interest: **State of Health (SOH)**, the current condition of a battery relative to its initial state, and **Remaining Useful Life (RUL)**, the operating life remaining before a defined degradation limit is reached.

Machine learning (ML) has become a leading alternative to laboratory-intensive estimation methods, learning the relationship between measurable signals (voltage, current, temperature, charging behaviour, cycle count) and battery degradation. A key concern, however, is that models built on clean laboratory data may degrade when inputs are noisy, sparsely sampled, incomplete, or limited to partial charging intervals, as is typical of real vehicle data. This review surveys the key literature and identifies the gap addressed by the proposed project.

2. Literature Review

2.1 Early-cycle lifetime prediction

Severson et al. (2019) trained models on 124 commercial lithium-iron-phosphate/graphite cells, using voltage-curve features from early cycles to predict cycle life. With the first 100 cycles, the best quantitative model reached a 9.1% test error, and classification using only the first five cycles reached 4.9%. The work established that early measurements carry enough information to forecast lifetime, but it relied on controlled laboratory data and did not examine the effect of degraded inputs.

2.2 ML pipelines for SOH estimation

Roman et al. (2021) built a pipeline using 179 cells under varied conditions, extracting 30 features from segments of charging voltage and current curves. After feature selection and comparison of parametric and non-parametric models, the best model achieved an RMSE of about 0.45% under fast charging, with confidence intervals on predictions. The study shows the value of engineered features and uncertainty estimation, but it does not systematically test performance under progressive noise, missing values, reduced sampling, or incomplete charging.

2.3 Standardisation and benchmarking

Zhang et al. (2024) introduced **BatteryML**, an open-source platform that unifies data preprocessing, feature extraction, and model implementation to address inconsistent formats and evaluation practices. Tan et al. (2025) introduced **BatteryLife**, a large benchmark addressing the small size, limited diversity, and narrow cell types of earlier datasets. Both improve reproducibility and generalisation testing, yet neither examines how much input degradation a model can tolerate.

2.4 Bridging the laboratory-to-field gap

Wang et al. (2024) downgraded laboratory data to approximate fleet-level operational data, extracting features from partial data ranges. They reported strong SOH estimation under these conditions and found that voltage-based and some mechanism-based features remained useful. This is the closest prior work to the proposed project, but it uses a single degradation configuration rather than a sweep across levels.

Lanubile et al. (2024) proposed a domain-knowledge-guided framework with five health indicators computable from partial charging profiles and real-world driving data. Models using power-autocorrelation and energy-based features achieved maximum absolute percentage errors of roughly 1.5–2.5%, even with partially missing data. The emphasis is on indicator design, not on measuring robustness across controlled degradation levels.

Liu et al. (2025) analysed three years of operational data from 300 EVs and developed a multi-modal deep-learning framework for SOH estimation. They confirm that laboratory datasets do not fully represent field conditions, but large-scale fleet data demands substantial preprocessing, computation, and domain expertise, limiting its practicality for smaller studies.

2.5 Explainability and validation reliability

Paneru et al. (2024) applied ensemble and hybrid ML models to RUL prediction with SHAP-based interpretation, showing that models can reveal which variables drive predictions. Explainability is valuable but is treated here as an optional extension.

Barua et al. (2026) found that random-split validation on NASA battery data gives substantially more optimistic results than leave-one-cell-out testing, because cycles from the same cell appear in both training and test sets. This motivates cross-battery validation in the present work.

3. Comparative Summary

Study	Focus	Contribution	Limitation for this project
Severson et al. (2019)	Cycle-life prediction	Early-cycle voltage features predict lifetime	Controlled laboratory data only
Roman et al. (2021)	SOH estimation	Strong pipeline with confidence intervals	No systematic degradation analysis
Zhang et al. (2024)	Standardisation	Unified preprocessing, features, models	Not focused on data-quality tolerance
Wang et al. (2024)	Downgraded lab data	Addresses lab-to-field gap directly	Single configuration, no multi-level sweep
Lanubile et al. (2024)	Real-world SOH	Indicators robust to partial data	Robustness not measured across controlled levels
Paneru et al. (2024)	RUL and explainability	Prediction with SHAP interpretation	Robustness to degraded data not central
Tan et al. (2025)	Benchmarking	Larger, more diverse dataset	Benchmarking, not degradation tolerance
Liu et al. (2025)	Real-world EV SOH	Evidence of lab-to-field differences	Requires large-scale fleet data
Barua et al. (2026)	Validation reliability	Exposes inflation from cell overlap	Concerned with leakage, not data quality

4. Key Themes

- **Growing use of ML.** Regression, tree-based, ensemble, and deep-learning models are all used for SOH and RUL estimation.
- **Importance of features.** Voltage, current, temperature, charging curves, and derived health indicators consistently carry degradation information.
- **Shift toward real-world data.** Recent work increasingly uses operational data, partial charging profiles, and downgraded laboratory measurements.
- **Concern about generalisation.** Strong results on conventional splits do not guarantee performance on unseen batteries.
- **Practical and interpretable models.** Uncertainty estimates, explainability, and physically meaningful indicators are gaining attention.

5. Research Gap

Existing studies show that ML can predict battery health well on laboratory or structured data. Real EV data, however, is often noisy, incomplete, sparsely sampled, and limited to partial charge or discharge windows. Although some studies use downgraded or real-world-like data, there is no simple, reproducible, systematic evaluation of how prediction quality changes across multiple levels of data degradation.

The question is therefore no longer whether ML can predict battery health, but:

> ***How much data quality does an ML battery-health predictor need before its predictions become unreliable?***

6. Proposed Approach

The project will use public battery-ageing datasets as a clean baseline and apply controlled degradation to the test data across four factors:

1. **Reduced sampling frequency**, simulating slower battery-management-system logging.
2. **Sensor noise and rounding**, simulating lower-precision sensors.
3. **Missing data**, simulating dropped measurements and communication gaps.
4. **Incomplete charging information**, simulating partial charging profiles.

Since the literature does not indicate that deep architectures are needed to answer this question, the study will use interpretable models: a cycle-count-only baseline, Ridge Regression, Random Forest, and XGBoost/Gradient Boosting. SOH and RUL will be predicted from voltage, current, temperature, charging, and cycle features, and evaluated with **leave-one-battery-out validation** to avoid the inflation reported by Barua et al. (2026).

Expected contributions

- A quantified relationship between data quality and prediction error.
- The minimum data quality needed to stay within a predefined error tolerance.
- A comparison of model robustness under degradation.

- Identification of feature groups that remain informative under noisy or incomplete data.
- A small Streamlit application for exploring the effect of data quality on predicted SOH and RUL.

7. Conclusion

The literature shows a clear progression: from early-cycle lifetime prediction, to engineered-feature SOH pipelines with uncertainty estimates, to standardised benchmarks, and most recently to real-world and partial-data settings. Each study addresses part of the problem, but none provides a systematic, multi-level assessment of robustness to degraded sensor data. The proposed project fills this gap with a controlled data-quality stress test using public datasets, classical ML models, and cross-battery validation, evaluating robustness rather than accuracy alone.

References

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